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3 Complex Analysis 2023
 A/Prof Elena Berdysheva
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Final Exam
 13 November 2023



Instructions:

- This question paper consists of **7 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 60 points = 100% (63 points available).

Problem	1	2	3	4	5	6	Σ	Mark
Maximum Points	12	8	10	13	10	10	60	100%
Result								

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Problem 1

(12 points)

- (a) Find the set of all $z \in \mathbb{C}$ where the function $f(z) = z \operatorname{Re} z$ is differentiable.
- (b) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{z^{2n}}{\sqrt{n}}$.

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Problem 2

(8 points)

Calculate the integral $\int_{\gamma_k} \frac{e^{z^2+1}}{(z-i)^2} dz$, $k = 1, 2$, where

(a) $\gamma_1(t) = 2 + e^{it}$, $t \in [0, 2\pi]$,

(b) $\gamma_2(t) = i + 2e^{it}$, $t \in [0, 2\pi]$.

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Problem 3

(10 points)

Assume that the functions f and g are holomorphic in the open disc $D(z_0; R) = \{z \in \mathbb{C} : |z| < R\}$, f has a zero of order $n \in \mathbb{N} \cup \{0\}$ at z_0 , and g has a zero of order $m > n$ at z_0 . Prove that $\frac{f}{g}$ has a pole at z_0 , and determine the order of this pole. Give detailed justifications.

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Problem 4

(13 points)

- (a) Determine the singularities of the function $f(z) = \frac{1}{z^3 - 8}$ in the closed upper half-plane $\{z \in \mathbb{C} : \text{Im } z \geq 0\}$, and calculate the residues at those singularities.
- (b) Give a definition of P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 - 8} dx$.
- (c) Calculate P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 - 8} dx$.

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Problem 5

(10 points)

Determine the Laurent series of the function $f(z) = \frac{1}{9 + z^2}$ in the given region. Give an explicit formula for the coefficients a_n , $n \in \mathbb{Z}$, of the Laurent series.

(a) $\{z \in \mathbb{C} : |z| < 3\}$,

(b) $\{z \in \mathbb{C} : |z| > 3\}$.

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Problem 6

(10 points)

How many zeros does the polynomial $p(z) = 2z^4 + 2z^3 - 5z + 1$ have in the disc $\{z \in \mathbb{C} : |z| < 2\}$?